Elastic vs Inelastic Collisions: Momentum and Energy

Momentum & Energy: Elastic and Inelastic Collisions · oPhysics

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics

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Simulation: https://ophysics.com/e2.html
Curriculum links: momentum · motion

Learning intentions

- I can apply conservation of momentum to a one-dimensional collision.
- I can distinguish elastic from inelastic collisions by comparing kinetic energy before and after.
- I can explain why momentum is always conserved in a collision but kinetic energy is not.

Getting started

Open the simulation with two boxes on a frictionless surface. Set the elasticity slider to 1 (perfectly elastic) first, then set the masses and initial velocities so the boxes are moving toward each other.

Tasks

1. With equal masses and elasticity = 1 (perfectly elastic), record the velocities before and after the collision. What do you notice about the two objects' velocities?

2. Set elasticity = 0 (perfectly inelastic) with the same masses and initial velocities. Calculate the expected combined velocity using conservation of momentum, then check it against the simulation.

Working:

3. Keeping the masses and initial velocities constant, run the collision at four different elasticity values and record the total kinetic energy before and after each.

Elasticity	KE before	KE after

4. Using your table from Task 3, calculate the percentage of kinetic energy lost at each elasticity value. Does momentum stay constant across all four trials, even though kinetic energy does not?

Working:

Extension (optional)

Explain, in terms of energy transformation, why momentum is conserved in every collision but kinetic energy is only conserved in perfectly elastic ones.